

GGL90 Validation Report

MITgcm vs Python Port Comparison (single-step replay)

Generated: 2026-09-27 05:56:48

MITgcm Reference:

File: mitgcm_ggl90_outputs_global_ocean_90x40x15_idemix_10.nc
UUID: d38655e7-bb98-4d5d-91e2-aac0014d029a

Python Port Output:

File: python_ggl90_outputs_global_ocean_90x40x15_idemix_10.nc
UUID: d38655e7-bb98-4d5d-91e2-aac0014d029a

Provenance: ☐ UUIDs match

Timesteps: 10

Grid: 90 × 40 × 15

Fields compared: visc_az, diff_kz, mixing_length, tke_after
(GGL90 single-step replay: Python fed MITgcm's own captured
tke_before(t); only the one-step update is compared, isolating
agreement on the physics formula from trajectory drift.)

GGL90 Field Statistics

Wet-cell comparison (NaN-excluded below-seafloor/above-ice-shelf cells)

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visc_az  (Vertical eddy viscosity (K_m), m^2/s)
          Status: □ EXCELLENT
Median |diff|: 2.0660e-17  P95 |diff|: 3.6910e-08  Max |diff|: 5.5864e+00
Median rel:   7.2575e-06   P95 rel:   7.5409e-04   Max rel:   2.1363e-01
          Fraction >1% rel error: 0.0377%
Points compared: 485840 (excluded 54160 NaN cells)

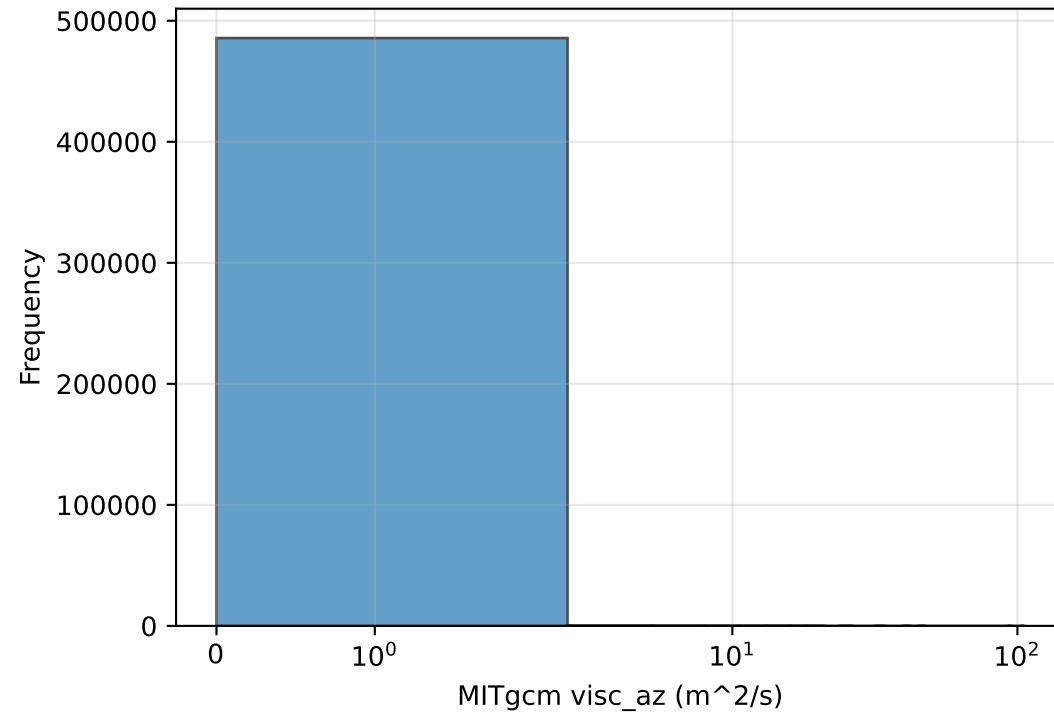
diff_kz  (Vertical eddy diffusivity (K_h), m^2/s)
          Status: △ CHECK
Median |diff|: 3.6387e-14  P95 |diff|: 3.8333e-05  Max |diff|: 2.7465e+00
Median rel:   1.4845e-02   P95 rel:   9.0004e-01   Max rel:   8.9998e+00
          Fraction >1% rel error: 50.9336%
Points compared: 485840 (excluded 54160 NaN cells)

mixing_length (GGL90 mixing length (L), m)
          Status: □ EXCELLENT
Median |diff|: 9.4093e-09  P95 |diff|: 4.3234e-04  Max |diff|: 1.5495e+01
Median rel:   3.4771e-05   P95 rel:   7.9336e-04  Max rel:   2.1363e-01
          Fraction >1% rel error: 0.0424%
Points compared: 485840 (excluded 54160 NaN cells)

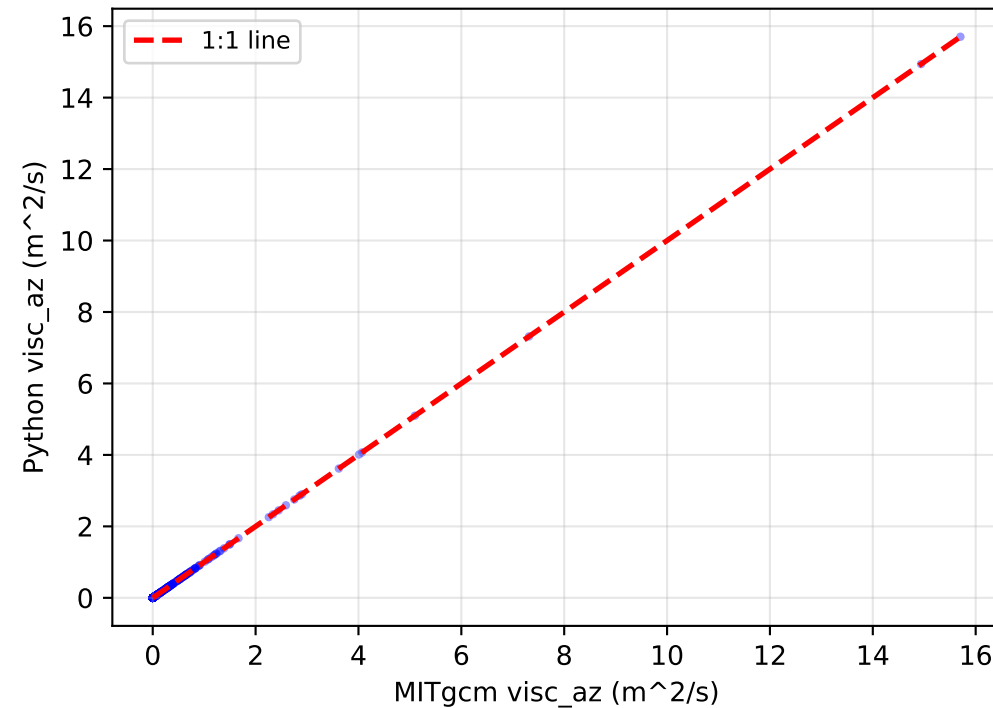
tke_after (TKE at t+1 (single-step replay), m^2/s^2)
          Status: △ CHECK
Median |diff|: 1.2131e-10  P95 |diff|: 1.6305e-06  Max |diff|: 4.4594e+02
Median rel:   9.8692e-01   P95 rel:   1.0000e+00  Max rel:   3.4633e+07
          Fraction >1% rel error: 54.4449%
Points compared: 485840 (excluded 54160 NaN cells)
```

visc_az (Vertical eddy viscosity (K_m)) -- MITgcm vs Python

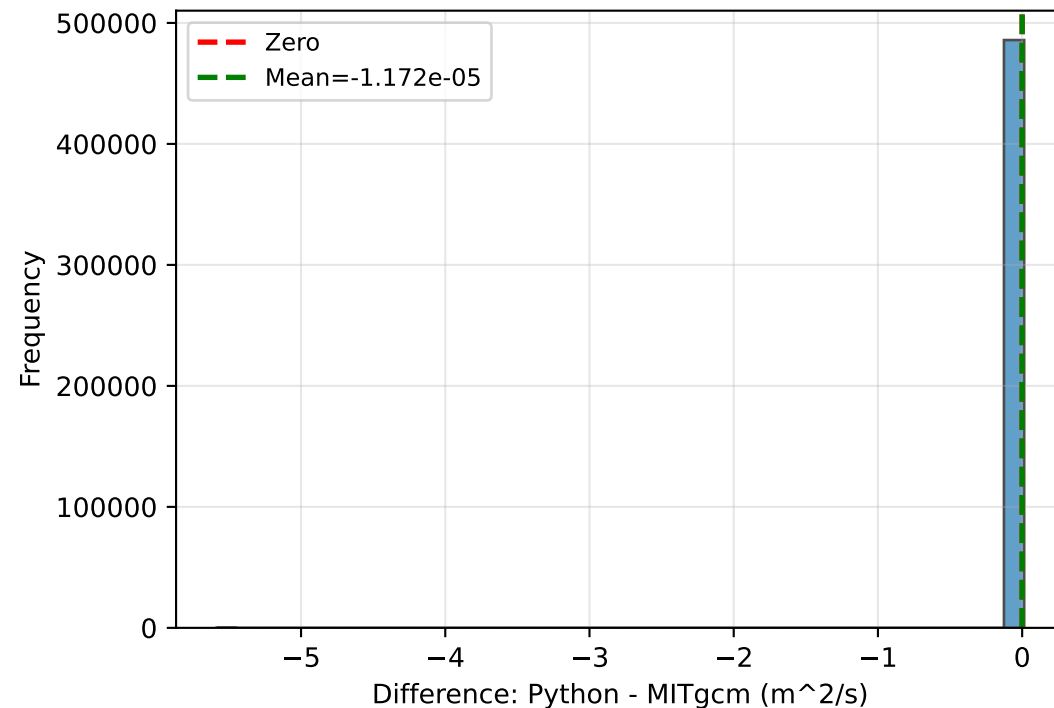
MITgcm Distribution



Scatter Plot (20000 of 485840 pts)



Difference Distribution



visc_az Statistics:

Median |diff|: 2.0660e-17
P95 |diff|: 3.6910e-08
Max |diff|: 5.5864e+00

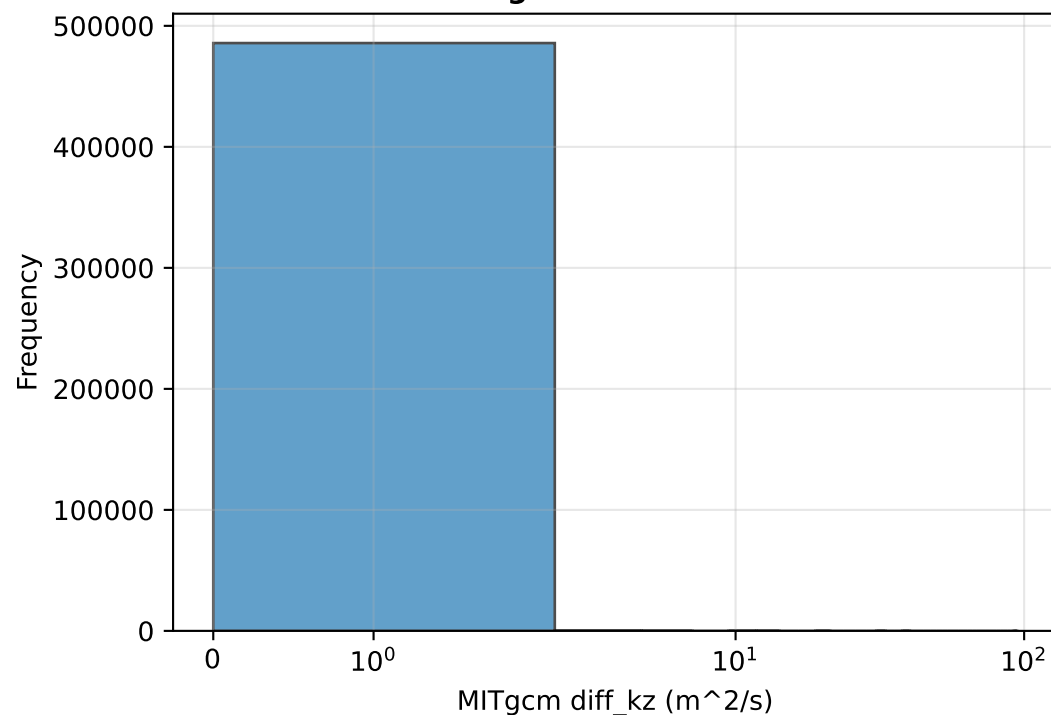
Median rel: 7.2575e-06
P95 rel: 7.5409e-04
Max rel: 2.1363e-01

Fraction >1% rel: 0.0377%
Points compared: 485840
NaN-excluded: 54160

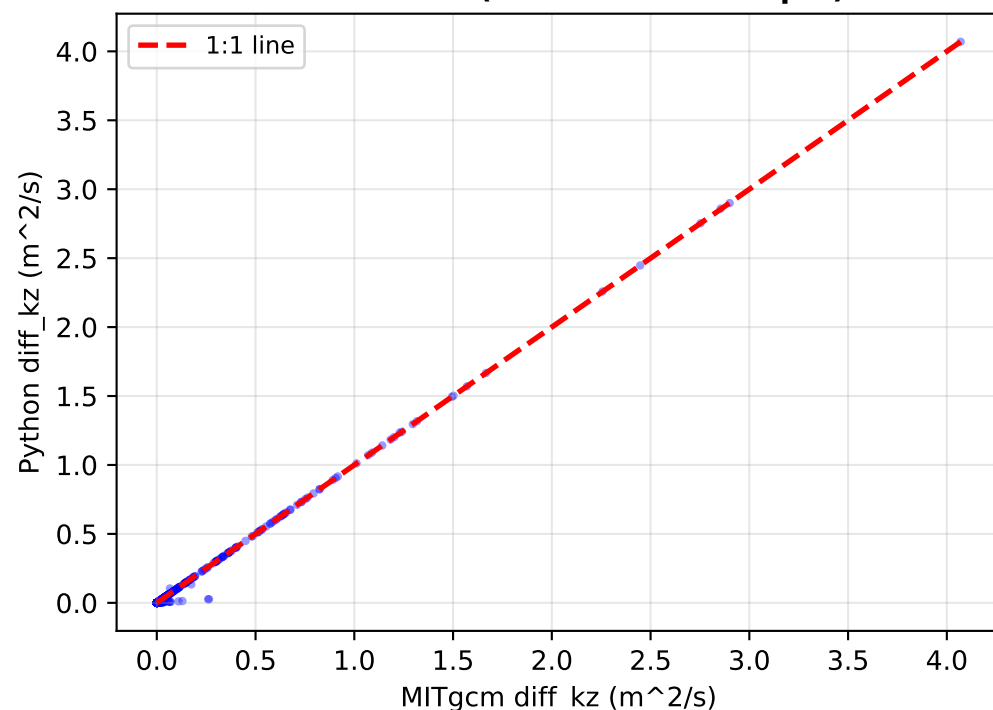
Assessment:
EXCELLENT

diff_kz (Vertical eddy diffusivity (K_h)) -- MITgcm vs Python

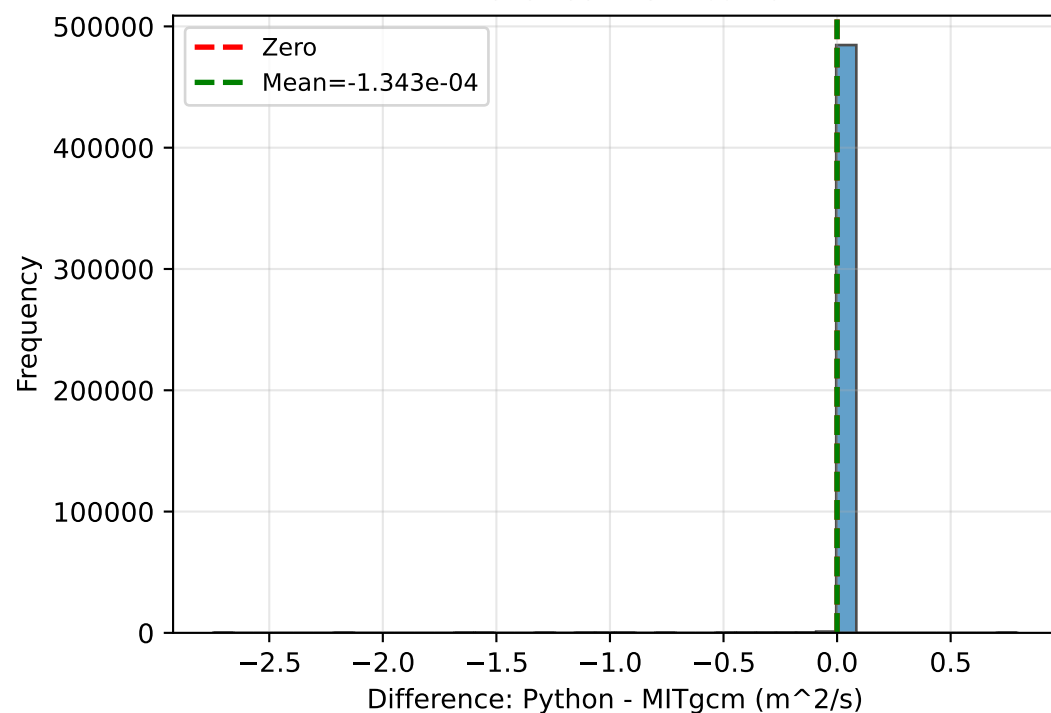
MITgcm Distribution



Scatter Plot (20000 of 485840 pts)



Difference Distribution



diff_kz Statistics:

Median |diff|: 3.6387e-14
P95 |diff|: 3.8333e-05
Max |diff|: 2.7465e+00

Median rel: 1.4845e-02
P95 rel: 9.0004e-01
Max rel: 8.9998e+00

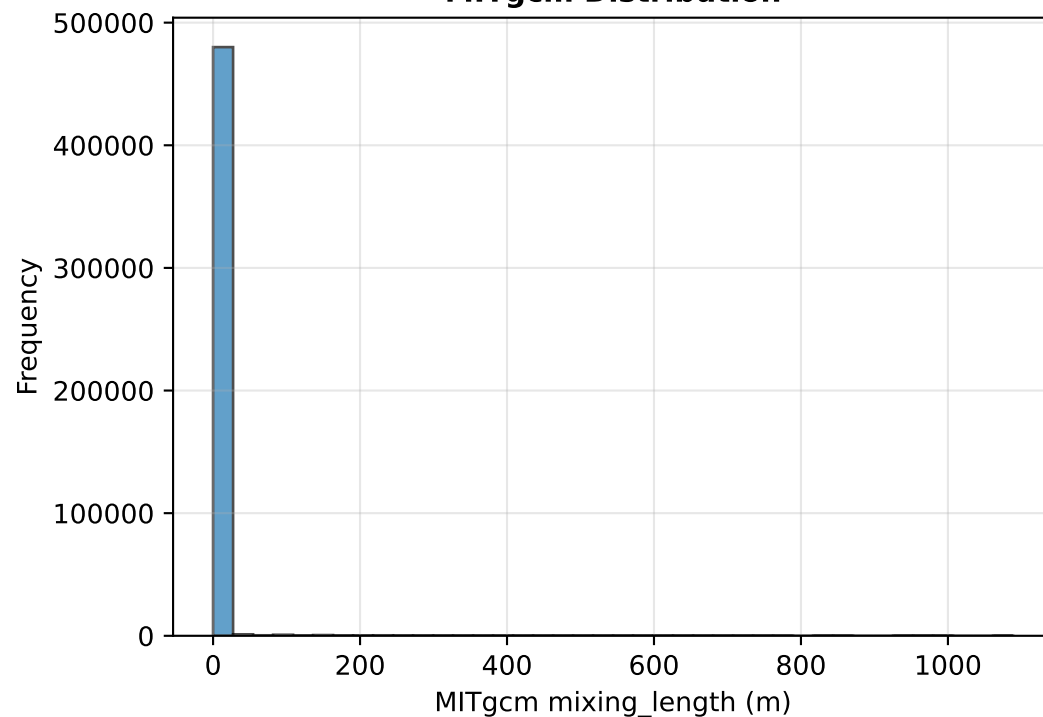
Fraction >1% rel: 50.9336%
Points compared: 485840
NaN-excluded: 54160

Assessment:

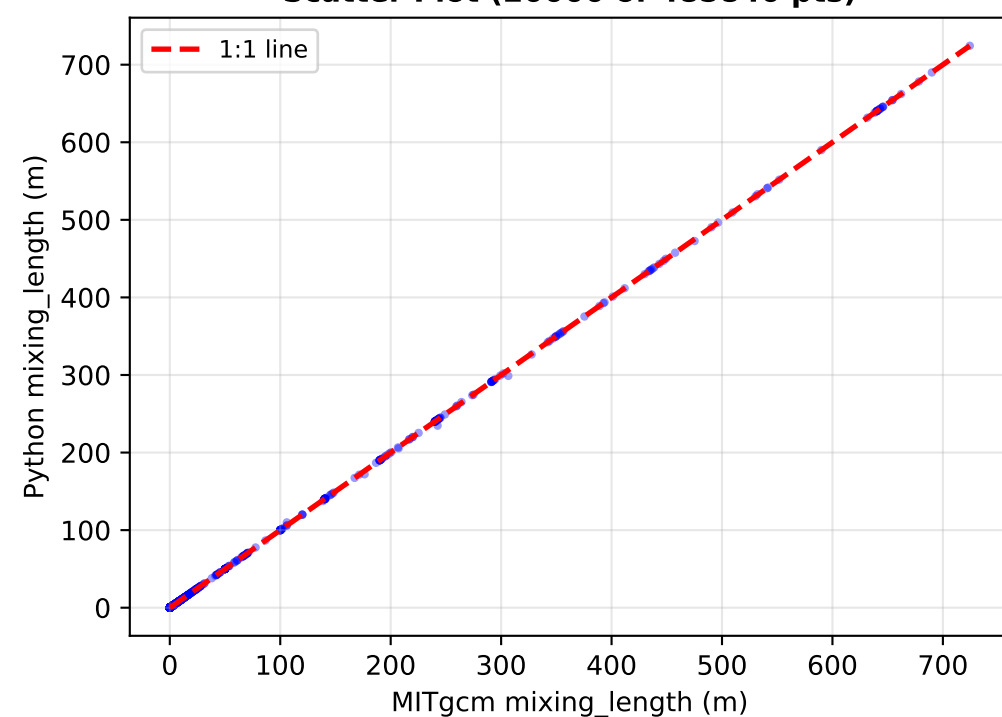
△ CHECK

mixing_length (GGL90 mixing length (L)) -- MITgcm vs Python

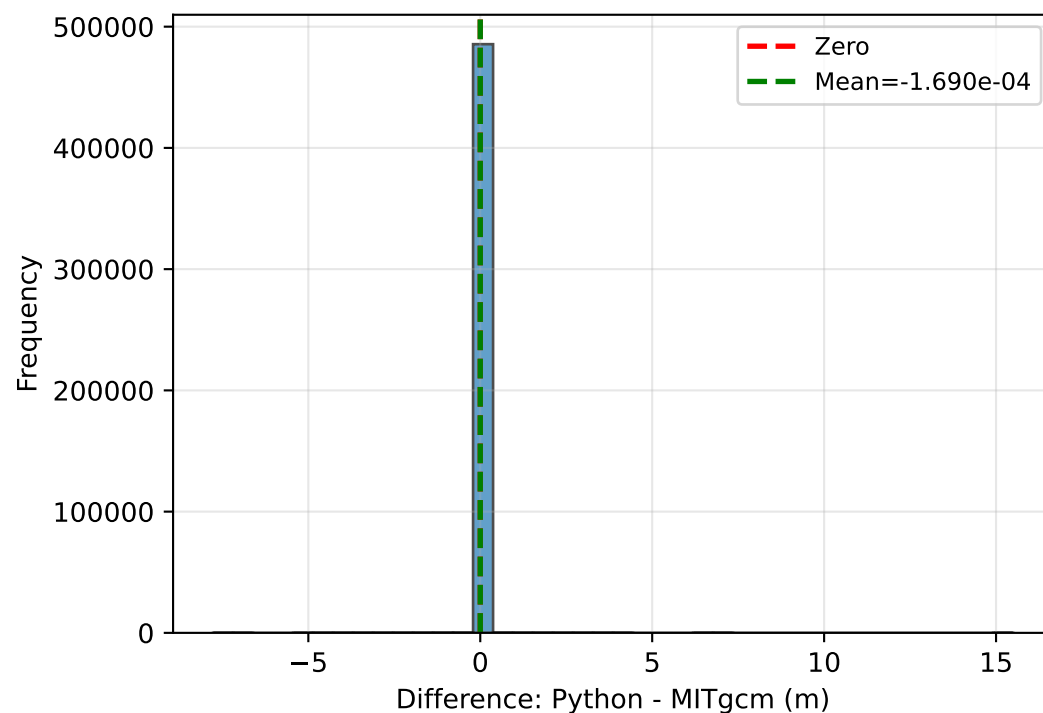
MITgcm Distribution



Scatter Plot (20000 of 485840 pts)



Difference Distribution



mixing_length Statistics:

Median |diff|: 9.4093×10^{-9}
P95 |diff|: 4.3234×10^{-4}
Max |diff|: 1.5495×10^1

Median rel: 3.4771×10^{-5}
P95 rel: 7.9336×10^{-4}
Max rel: 2.1363×10^1

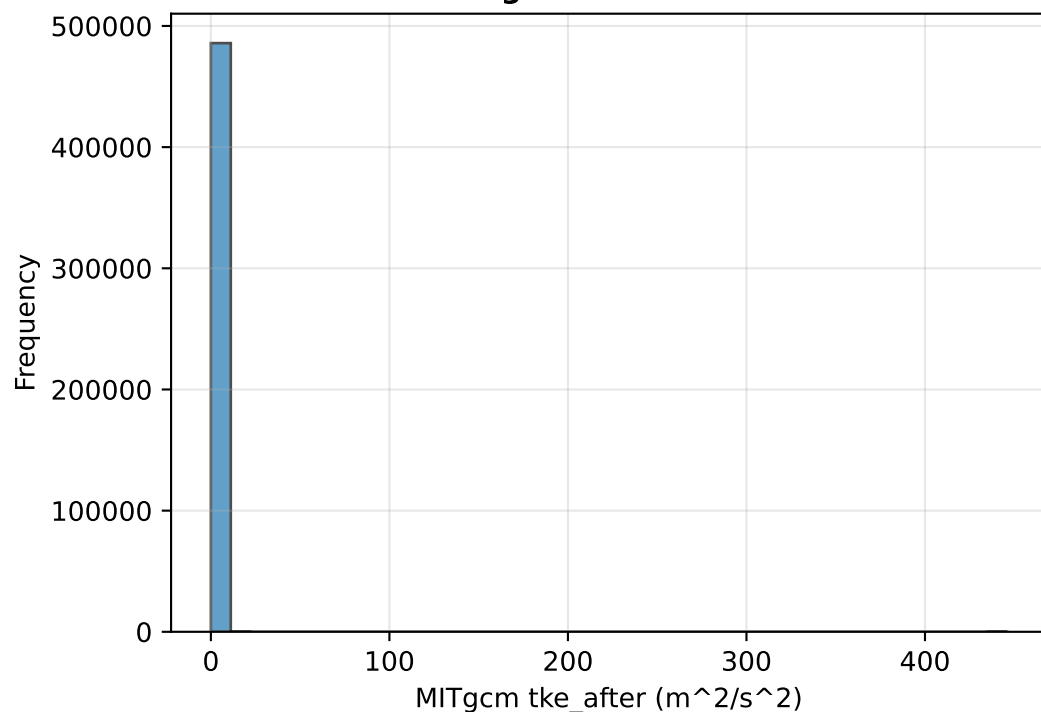
Fraction >1% rel: 0.0424%
Points compared: 485840
NaN-excluded: 54160

Assessment:

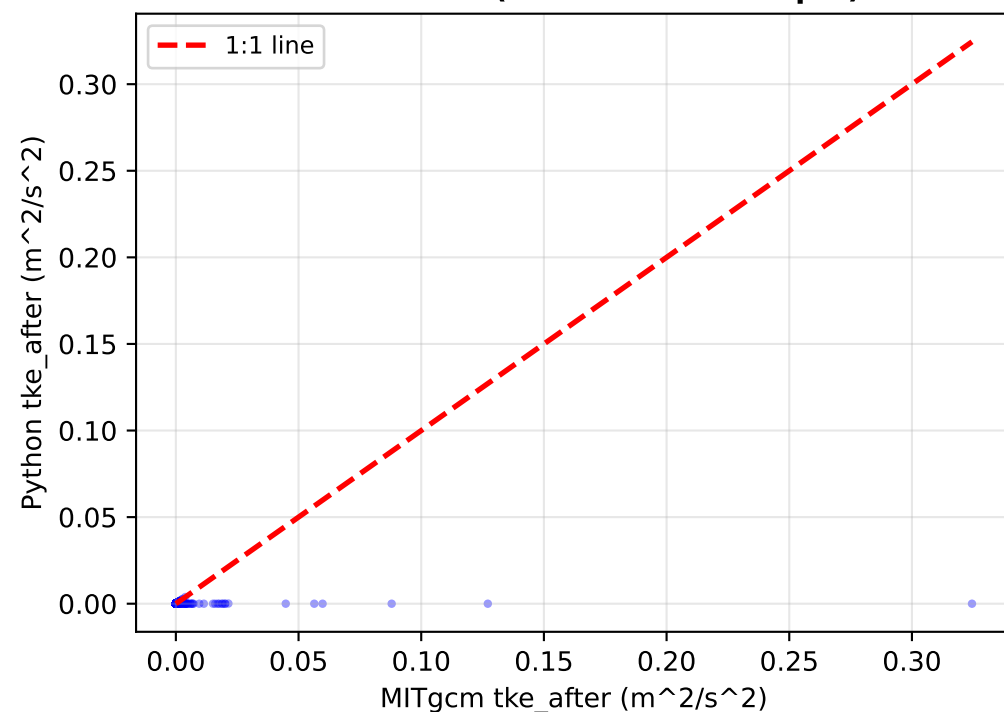
EXCELLENT

tke_after (TKE at t+1 (single-step replay)) -- MITgcm vs Python

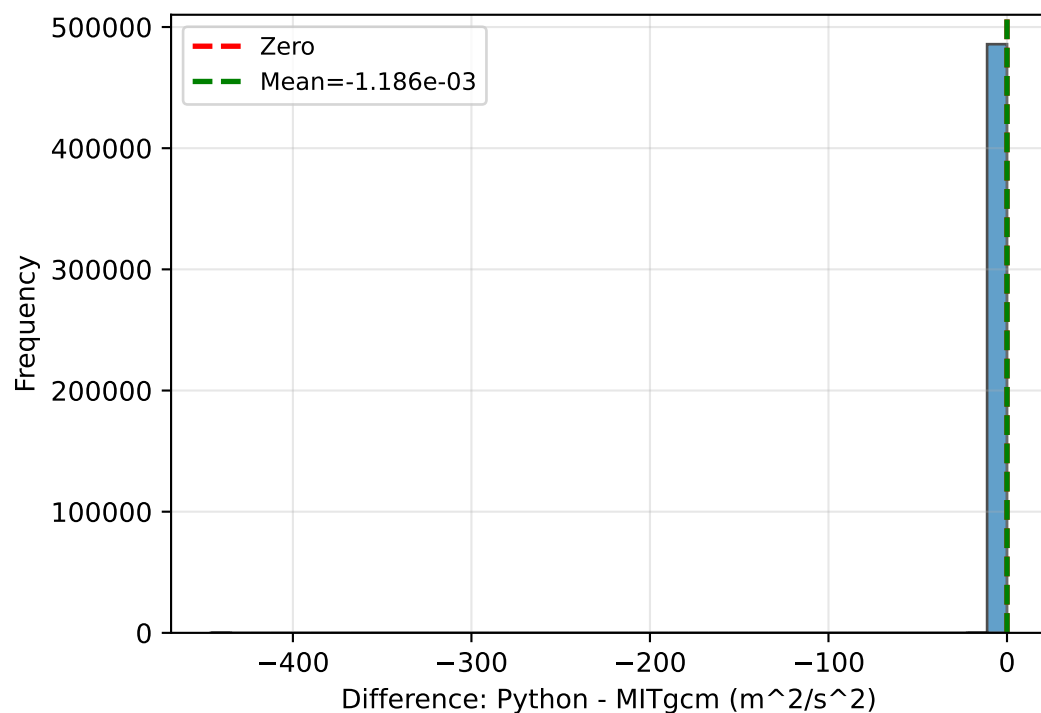
MITgcm Distribution



Scatter Plot (20000 of 485840 pts)



Difference Distribution



tke_after Statistics:

Median |diff|: 1.2131e-10
P95 |diff|: 1.6305e-06
Max |diff|: 4.4594e+02

Median rel: 9.8692e-01
P95 rel: 1.0000e+00
Max rel: 3.4633e+07

Fraction >1% rel: 54.4449%
Points compared: 485840
NaN-excluded: 54160

Assessment:

△ CHECK

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: Δ ☐ NEEDS REVIEW

COMPONENT RESULTS:

visc_az: ☐ EXCELLENT
Median rel error: 0.0007% Max |diff|: 5.5864e+00
Fraction >1% rel: 0.0377%

diff_kz: Δ ☐ CHECK
Median rel error: 1.4845% Max |diff|: 2.7465e+00
Fraction >1% rel: 50.9336%

mixing_length: ☐ EXCELLENT
Median rel error: 0.0035% Max |diff|: 1.5495e+01
Fraction >1% rel: 0.0424%

tke_after: Δ ☐ CHECK
Median rel error: 98.6917% Max |diff|: 4.4594e+02
Fraction >1% rel: 54.4449%

PROVENANCE:

UUID Match: ☐ Yes
MITgcm UUID: d38655e7-bb98-4d5d-91e2-aac0014d029a
Python UUID: d38655e7-bb98-4d5d-91e2-aac0014d029a

ACCEPTANCE CRITERIA (mirrors generate_kpp_validation_report.py's own):

☐ EXCELLENT: median rel error < 0.1%
☐ GOOD: median rel error < 1.0%
 Δ ☐ CHECK: median rel error >= 1.0% -- may be a real, already-characterized gap (see closed_issues.md), not necessarily a new defect; this report states the number, not the cause.